Characterizing Galois extensions

Example: $M = \dfrac{\mathbb{Q}[x]}{(x^3-2)} \ni \alpha = x + (x^3-2)$

$$\mathrm{Emb}_{\mathbb{Q}}(M, \mathbb{C}) \simeq \{\alpha \in \mathbb{C} : \alpha^3 = 2\}$$
$$= \{\sqrt[3]{2},\ \zeta_3\sqrt[3]{2},\ \zeta_3^2\sqrt[3]{2}\}$$

$$\zeta_3 = e^{2\pi i/3}$$

$$\begin{tikzcd} \mathbb{Q}[x] \arrow[d] \arrow[dr, "ev_{\sqrt[3]{2}}"] & \\ M \arrow[r, "\tau_1"] & \mathbb{C} \end{tikzcd}$$

$\tau_1 : M \longrightarrow \mathbb{C}$

$\alpha \longmapsto \sqrt[3]{2}$

$\tau_2 : M \longrightarrow \mathbb{C}$

$\alpha \longmapsto \zeta_3\sqrt[3]{2}$

$\tau_3 : M \longrightarrow \mathbb{C}$

$\alpha \longmapsto \zeta_3^2\sqrt[3]{2}$

$\tau_1(M) = \mathbb{Q}(\sqrt[3]{2}) \leq \mathbb{C}$

$\tau_2(M) = \mathbb{Q}(\zeta_3\sqrt[3]{2}) \leq \mathbb{C}$

$\tau_3(M) = \mathbb{Q}(\zeta_3^2\sqrt[3]{2}) \leq \mathbb{C}$

these are distinct!

Consider

$$N = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) \leq \mathbb{C}$$

$$\begin{tikzcd}
N = (\mathbb{Q}(\sqrt[3]{2}))(\zeta_3) \arrow[d, dash, "\text{deg } 2" ', "x^2+x+1"] \\
\mathbb{Q}(\sqrt[3]{2}) \arrow[d, dash, "\text{deg } 3" ', "x^3-2"] \\
\mathbb{Q}
\end{tikzcd}$$

$$[N : \mathbb{Q}] = 6$$

$$\begin{tikzcd}
 & N \arrow[dl, dash, "\text{deg } 2" '] \arrow[d, dash, "\text{deg } 2" '] \arrow[dr, dash, "\text{deg } 2"] \arrow[drr, dash, "\text{deg } 3"] & & \\
\mathbb{Q}(\zeta_3 \sqrt[3]{2}) \arrow[dr, dash, "\text{deg } 3" '] & \mathbb{Q}(\sqrt[3]{2}) \arrow[d, dash, "\text{deg } 3"] & \mathbb{Q}(\zeta_3^2 \sqrt[3]{2}) \arrow[dl, dash, "\text{deg } 3"] & \mathbb{Q}(\zeta_3) \arrow[dll, dash, "\text{deg } 2"] \\
 & \mathbb{Q} & &
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{Q}(\sqrt[3]{2})(\rho_3) = N \arrow[r] \arrow[d, dash] & N \arrow[d, equal] \\
M \cong \mathbb{Q}(\sqrt[3]{2}) \arrow[r, "\tau_1 \text{ or } \tau_2 \text{ or } \tau_3"'] \arrow[d, dash] & N \\
\mathbb{Q} &
\end{tikzcd}$$

Fix $i = 1, 2, 3$

$$\left[\begin{array}{l} i=1: \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ i=2: \sqrt[3]{2} \mapsto \rho_3 \sqrt[3]{2} \\ i=3: \sqrt[3]{2} \mapsto \rho_3^2 \sqrt[3]{2} \end{array}\right]$$

Pick $i=2$: $\sqrt[3]{2} \mapsto \rho_3 \sqrt[3]{2}$

$$\begin{tikzcd}
N \arrow[r, dotted] \arrow[d, dash] & N \arrow[d, dash] \arrow[dl, dash] \\
M \arrow[r, "\cong"] & \tau_2(M) \\
\alpha \arrow[r, mapsto] & \tau_2(\alpha)
\end{tikzcd}$$

How many ways do I have to fill in this dotted arrow

Concretely: How many $\sigma : N \to N$ do we have s.t. $\forall \alpha \in M$,

$$\sigma(\alpha) = \tau_2(\alpha)$$

$$N = M(\zeta_3) = \frac{M[x]}{(x^2+x+1)}$$

$$\mathrm{Emb}_M(N, L) = \{\beta \in L : \beta^2+\beta+1 = 0\}$$

L/M: any extension

$$\mathrm{Emb}_M(N, N) \Rightarrow \{\zeta_3, \zeta_3^2\}$$

$\Rightarrow$ there are two $\sigma : N \to N$ with

$$\sigma(\alpha) = \tau_L(\alpha), \ \forall \alpha \in M$$

We can write down 6 elements

in $\mathrm{Emb}_{\mathbb{Q}}(N, N) = \mathrm{Aut}(N/\mathbb{Q})$

$$\left\{\begin{matrix} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{matrix}\right\}, \quad \left\{\begin{matrix} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \end{matrix}\right\} = \tau$$

$$\sigma = \left\{\begin{matrix} \sqrt[3]{2} \mapsto \zeta_3 \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{matrix}\right\}, \quad \left\{\begin{matrix} \sqrt[3]{2} \mapsto \zeta_3 \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \end{matrix}\right\}$$

$$\left\{\begin{matrix} \sqrt[3]{2} \mapsto \zeta_3^2 \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{matrix}\right\}, \quad \left\{\begin{matrix} \sqrt[3]{2} \mapsto \zeta_3^2 \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \end{matrix}\right\}$$

$N/\mathbb{Q}$ is Galois,

and $|\mathrm{Gal}(N/\mathbb{Q})| = 6$

<u>Compute:</u>

$$\sigma\circ\tau: \sqrt[3]{2} \xmapsto{\tau} \sqrt[3]{2} \xmapsto{\sigma} \zeta_3\sqrt[3]{2}$$

$$\zeta_3 \xmapsto{\tau} \zeta_3^2 \xmapsto{\sigma} \zeta_3^2$$

$$\neq$$

$$\tau\circ\sigma: \sqrt[3]{2} \xmapsto{\sigma} \zeta_3\sqrt[3]{2} \xmapsto{\tau} \zeta_3^2\sqrt[3]{2}$$

$$\tau(\zeta_3\sqrt[3]{2}) = \tau(\zeta_3)\tau(\sqrt[3]{2})$$

$$\zeta_3 \xmapsto{\sigma} \zeta_3 \xmapsto{\tau} \zeta_3^2$$

$$\Rightarrow \mathrm{Gal}(N/\mathbb{Q}) \cong D_6$$

$$
\begin{tikzcd}
 & \langle e \rangle \arrow[dl, dash, "\text{index } 2"'] \arrow[d, dash, "\text{index } 2"] \arrow[dr, dash, "\text{index } 2"] \arrow[drr, dash, "\text{index } 3"] & & \\
\langle \tau\sigma^2 \rangle \arrow[dr, dash, "\text{index} = 3"'] & \langle \tau \rangle \arrow[d, dash, "\text{index } 3"'] & \langle \tau\sigma \rangle \arrow[dl, dash, "\text{index } 3"] & \langle \sigma \rangle \arrow[dll, dash, "\text{index} = 2"] \\
 & D_6 & &
\end{tikzcd}
$$

Normal!

$$
\begin{tikzcd}
 & N \arrow[dl, dash, "\deg 2"'] \arrow[d, dash, "\deg 2"'] \arrow[dr, dash, "\deg 2"] \arrow[drr, dash, "\deg 3"] & & \\
\mathbb{Q}(\zeta_3^2 \sqrt[3]{2}) \arrow[dr, dash, "\deg 3"'] & \mathbb{Q}(\sqrt[3]{2}) \arrow[d, dash, "\deg 3"'] & \mathbb{Q}(\zeta_3 \sqrt[3]{2}) \arrow[dl, dash, "\deg 3"] & \mathbb{Q}(\zeta_3) \arrow[dll, dash, "\deg 2"] \\
 & \mathbb{Q} & &
\end{tikzcd}
$$

Galois!

Thm M/K: finite extension

TFAE:

(1) M is Galois over K

(2) For all subextensions $M/L/K$

$$|\mathrm{Emb}_K(L,M)| = [L:K]$$

(3) $\forall \alpha \in M$, with minimal poly. $m_\alpha(x) \in K[x]$ of deg d

$m_\alpha(x)$ has exactly d distinct zeros in M

(4) $f(x) \in K[x]$ irred. of deg d

If $f(x)$ admits one zero in M

then $f(x)$ has d distinct zeros in M

(5) There exists a separable irreducible polynomial $f(x) \in K[x]$ of deg d

s.t. (i) $f(x)$ has d distinct zeros in M
$\alpha_1, \dots, \alpha_d \in M$

(ii) $M = K(\alpha_1, \dots, \alpha_d)$